Predicting Economic Damage of Natural Disasters

Hyeonsu An, Max Niu

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```
# A tibble: 14 x 2
   disaster_type                 n
   <chr>                     <int>
 1 Animal incident               1
 2 Drought                     422
 3 Earthquake                  693
 4 Epidemic                    892
 5 Extreme temperature         568
 6 Flood                      4243
 7 Glacial lake outburst flood     8
 8 Impact                        1
 9 Infestation                  29
10 Mass movement (dry)          14
11 Mass movement (wet)         494
12 Storm                      2823
13 Volcanic activity           133
14 Wildfire                    338
```

1 Introduction

1.1 Background:

As global temperatures rise, climate-related disasters have increased in both frequency and severity over the past several decades (World Wild Life). Extreme weather events such as floods, droughts, hurricanes, etc. impose a huge human and real economic costs across the world. It is imperative that we understand the determinants of disaster severity (in terms of economic damages) for better policy planning and mitigative strategies.

The Emergency Events Database (EM-DAT), maintained by the Centre for Research on the Epidemiology of Disasters (CRED), keeps an internationally standardized data on natural diasters worldwide. The information includes disaster type, location, affected population, deaths, injuries, and economic damages.

Include discussion about work others have done related to your research question.

1.2 Research Question:

How do different disaster characteristics (such as geographic location, time of year, type, and total casualties incurred) predict the total economic damage caused by natural disasters in USD?

- Specifically, we examine whether disaster type, location, timing, duration, and measures of human impact (deaths, injuries, and affected population) are associated with higher reported economic losses.

1.3 Motivation:

We were first drawn to this research question because were intrigued by this data; we were amazed by the granularity and the surprising breadth of information in this data. However, we realized that doing research on this matter could have real, practical uses. Prior work, such as Noy (2009), has shown that country-level factors like institutional strength and trade openness shape how severely a disaster disrupts an economy — but this line of research focuses on macroeconomic outcomes rather than predicting damage from the observable characteristics of individual disaster events. Our project aims to fill this gap.

Identifying predictors of large economic damage is critical for both public and private institutions. Governments and international organizations allocate use resources towards mitigation and recovery efforts, yet the determinants of these disasters vary across event types and regions. The same applies to chase alpha. By analyzing global disaster data, this project aims to quantify which observable characteristics are most predictive of economic loss.

1.4 Hypothesis Reasoning:

Based on economic reasoning and relevant research, we expect that…

1. Disasters with higher number of affected individuals (absolute number, number of deaths and injuries) will have a positive relationship with economic damage.
2. Certain disaster types will incur systematically higher economic damage than that of others.

3. The level of damage dollars will defer based on countries because of their respective infrastructure that exists to mitigate damage.

1.5 Data Description

This data was obtained from the Emergency Events Database, otherwise known as EM-DAT, which is maintained by the Centre for Research on the Epidemiology of Disasters (CRED). The goal of the database is to compile standardized information on natural disasters that occur around the world. According to the information on metadata, an event is included in EM-DAT if one or more of the following criteria is met:

- 10+ people killed
- 100+ people affected
- Declaration of a state of emergency
- Solicitation of international assistance

Each observation is a different disaster that occurred around the world. Of each disaster, data on the number of people injured, affected, homeless, or killed, the total economic damages, type of disaster, exact location, start date and end date of the disaster were collected.

The data dictionary can be found here.

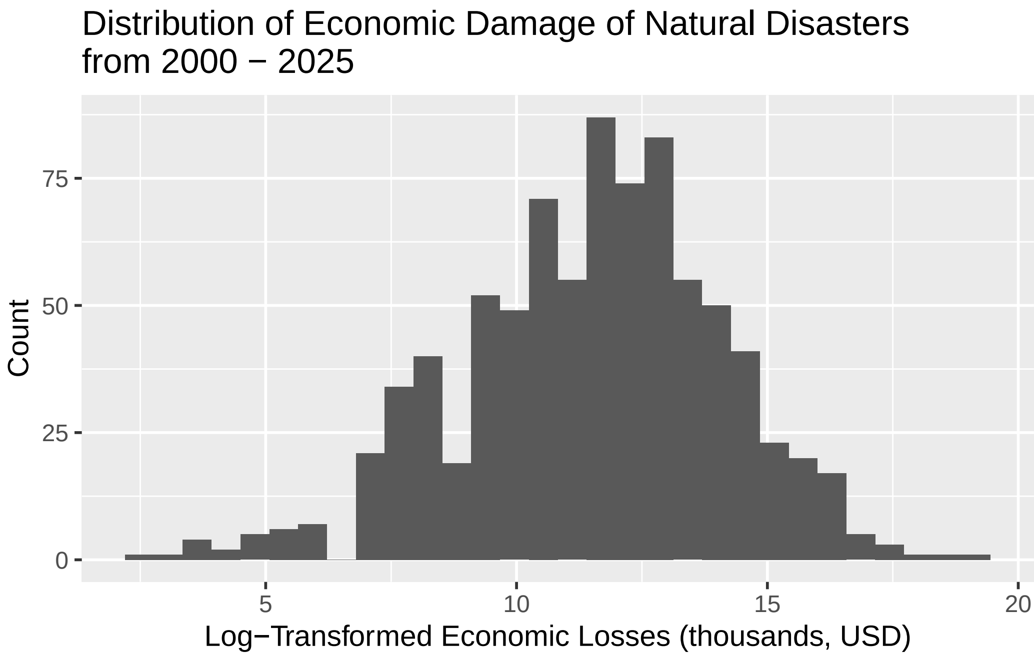
1.6 Data Cleaning

To clean the raw natural disaster dataset, variable names were standardized using clean_names(). Next, the dataset was reduced to only the variables relevant to the analysis. These included information on disaster type, location (country), event timing (start and end year, month, and day), and key impact measures such as total deaths, number injured, number affected, total affected, and total economic damage (in thousands of USD). Rows containing missing values were then removed using drop_na(). We attempted to make sure there was no relationship between the observations with missing values. Using the event timing data, a new variable duration_days was calculated as the difference between the end date and start date. An additional day was added to this value to ensure that the duration includes the start date.

We considered combining the levels of disaster_type with few observations into one level named "Other", but after deliberation, we decided to keep all the levels because although our cleaned data set had few observations for volcanic activity, extreme temperature, and mass movement (wet), in the raw data set, there are 338 observations for volcanic activity, 568 observations for extreme temperature, and 494 observations for mass movement (wet). This demonstrates that these natural disasters are common and thus, being able to predict the total economic damage associated with these disasters is important.

1.8 Exploratory Data Analysis

Response variable - log-transformed total economic damage in thousands of USD

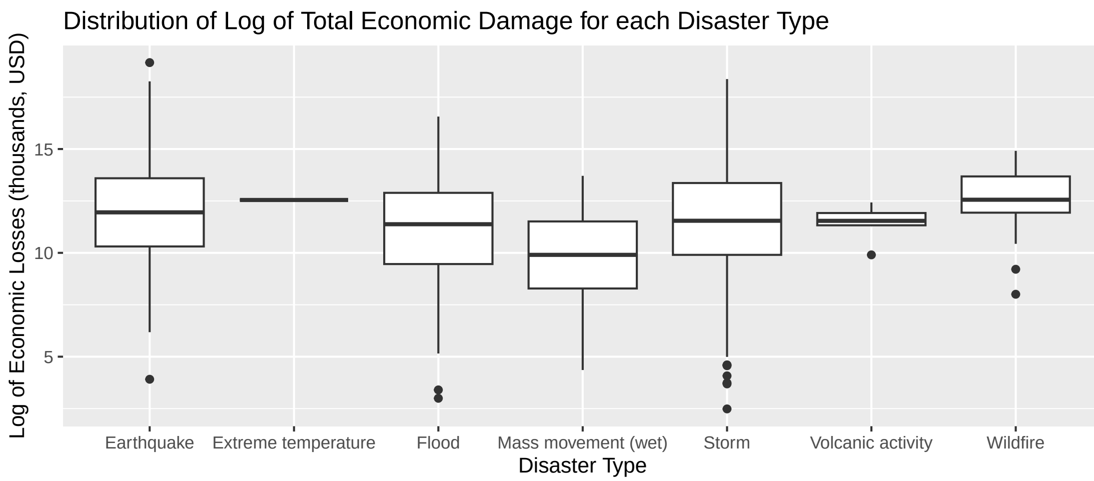


min	1st_qu	median	mean	3rd_qu	max	sd
2.485	9.903	11.604	11.493	13.235	19.163	2.606

We will be working with log-transformed total economic damage in thousands of USD, because without the log transformation, the 123 outliers skew the data heavily to the right. The distribution of the logged response variable is unimodal and symmetric. The center is best represented by the mean, which is $98,027,173.52 (e^11.493 * 1000), and the spread is best represented with the standard deviation, which is $13544.98 (e^2.606 * 1000). There are outliers present in our distribution.

Predictors of interests-

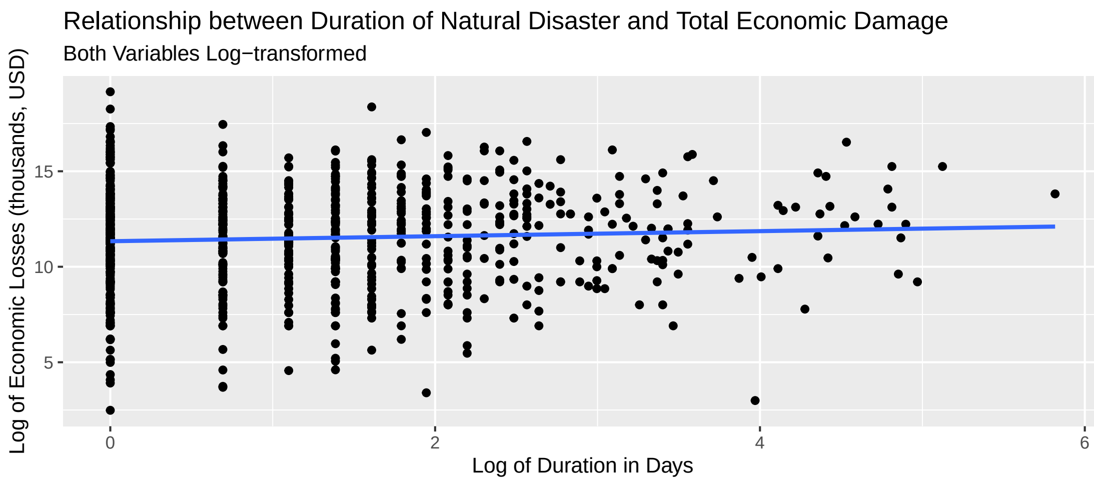
1. disaster_type - the type of disaster (categorical)



disaster_type	median	IQR
Earthquake	11.951	3.283
Extreme temperature	12.546	0.000
Flood	11.379	3.429
Mass movement (wet)	9.903	3.229
Storm	11.546	3.459
Volcanic activity	11.542	0.592
Wildfire	12.559	1.746

Wildfires have the highest median log of total economic damage (in thousands of dollars), which is 12.559, and thus, normal total economic damage, which would be 284,604.84 thousands of dollars (e^12.559). The median of earthquakes is slightly lower, and then the median of volcanic activities, floods, and storms, which all have similar values, slightly below the median of earthquakes. Wet mass movements have the lowest median (log of) total economic damage. Storms, floods, and earthquakes have the highest variability in the (log of) total economic damage.

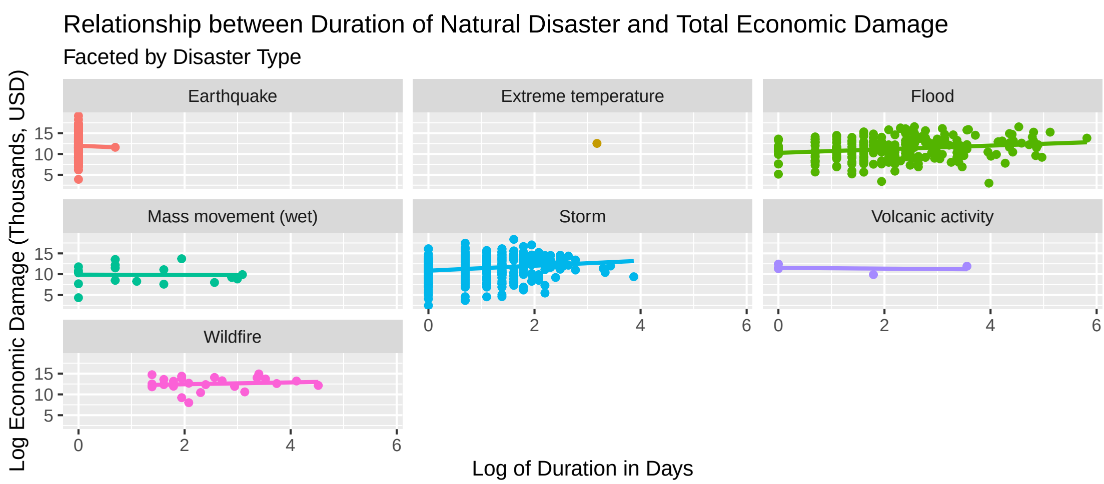
2. duration - duration in days of the disaster (quantitative)




```
# A tibble: 1 x 1
  r_squared
      <dbl>
1   0.00360
```

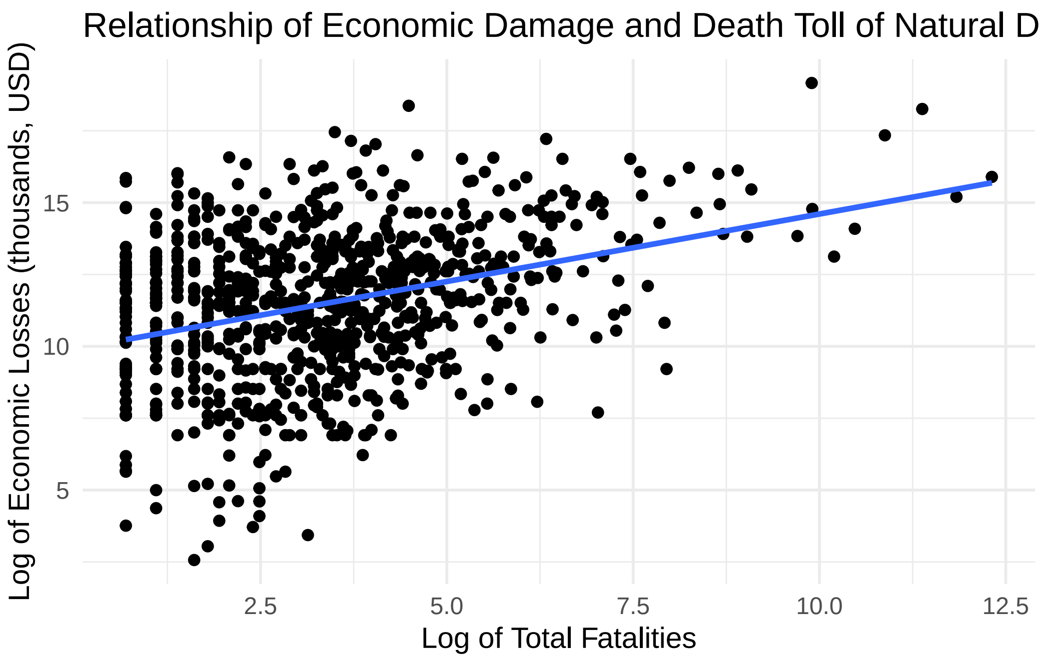
There seems to be a weak positive linear relationship between the log of duration in days with log of total economic damage (in thousands of dollars). The r-squared value is 0.003596655. There is a lot of variability in the response variable when the log of duration in days is low, but the variability decreases as the x value increases. There are outliers present throughout the graph.

Based off of our knowledge about how the duration of various disasters and how that should affect economic damage, we then decide to explore an interaction effect between log_duration_days and disaster_type.



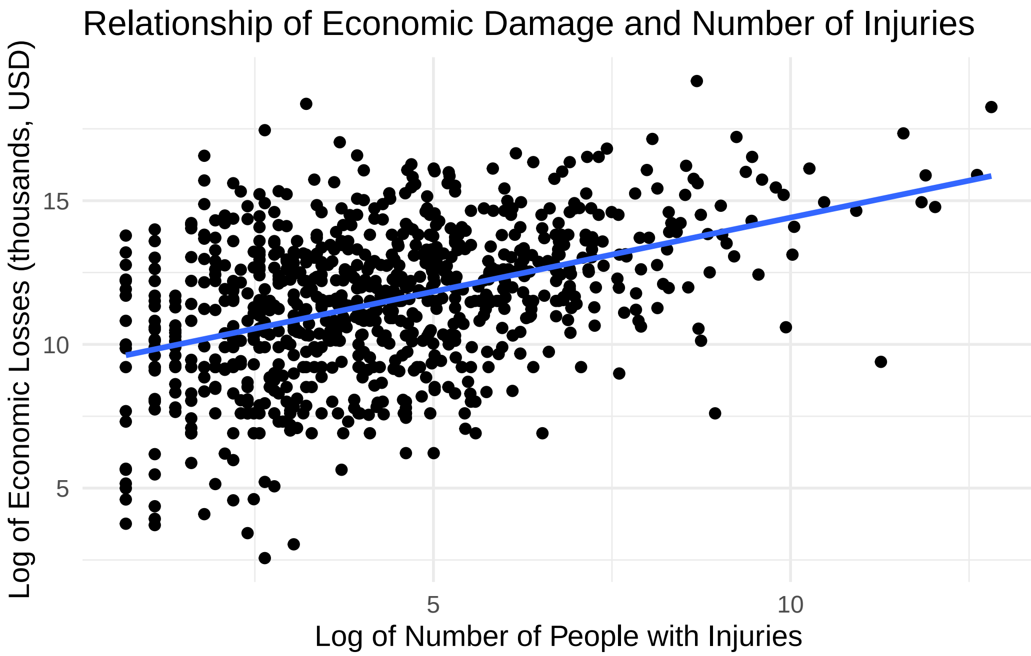
After visualizing the potential interaction effect between log duration of days and type of disaster, it is evident that an increase in the log duration in days has a stronger effect on log total economic damage for storms and floods, compared to wildfires, volcanic activities, and wet mass movements, which is shown by the slightly steeper positive slope. We will consider putting this interaction effect into our model because of this visualization.

3. total_deaths - Total fatalities (deceased and missing combined) (quantitative)



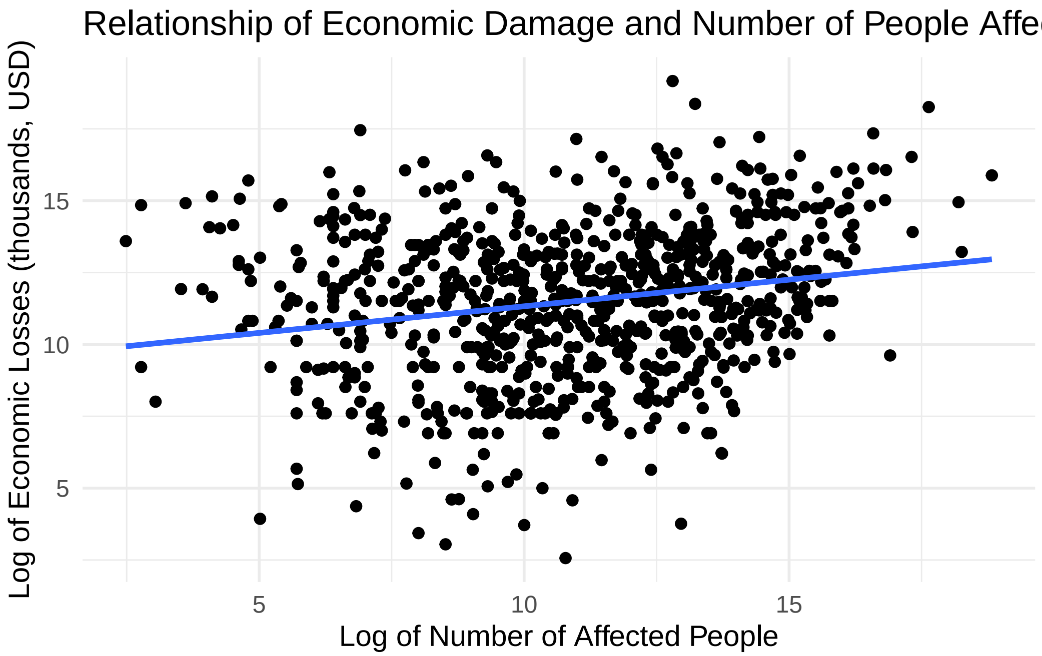
There is a weak positive linear relationship between log of total deaths and economic damage with an r-squared value of 0.116. Disasters with higher fatalities generally correspond to higher economic losses (USD). However, the relationship is noisy and dispersed, indicating that deaths alone may not be a strong predictor of damage and that further analysis needs to be done.

4. no_injured - Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster (quantitative)



There is a weak positive linear relationship between log of total injuries and economic damage with an r-squared value of 0.18. This relationship is similar to log_total_deaths. Although slightly less, the relationship is noisy and dispersed, indicating that log number of injuries may not be a strong predictor of damage and that further analysis needs to be done.

5. no_affected - Number of people requiring immediate assistance due to the disaster (quantitative)



There is a weak positive linear relationship between log of total injuries and economic damage with an r-squared value of 0.044. This relationship is similar to log_total_deaths and log_no_injured, but the r-squared value is lower. This means that the relationship is more noisy and dispersed and that there is a stronger relationship between log total deaths and log number of injuries than log of number of people affected with log of total economic damages. We will keep this in mind while finding the best model.

2 Methodology

2.1 Evaluating the interaction effect -

Based on our EDA, we were curious to see if a model with an interaction effect between log_duration_days and disaster_type would be a better model, so we conducted 10-fold cross validation to see which model performed better.

- Model 1 (without interaction): log(no_affected) + log_duration_days + disaster_type + log(total_deaths) + log(no_injured)
- Model 2 (with interaction): log_total_damage_000_us ~ log(no_affected) + **log_duration_days * disaster_type** + log(total_deaths) + log(no_injured)

```
# A tibble: 4 x 5
  model   .metric  mean     n std_err
  <chr>   <chr>   <dbl> <int>   <dbl>
1 Model 1 rmse    2.23     10  0.0287
2 Model 1 rsq     0.236    10  0.0310
3 Model 2 rmse    2.23     10  0.0287
4 Model 2 rsq     0.240    10  0.0307
```

After conducting cross-validation, we observed similar values for rmse and r-squared for model 1 and model 2. Because model 1 (the model without the interaction effect) is the more parsimonious model, we are going to move forward with it.

Model 1:

term	estimate	std.error	statistic	p.value
(Intercept)	8.898	0.440	20.239	0.000
log(no_affected)	0.007	0.039	0.185	0.853
log_duration_days	0.285	0.107	2.667	0.008
disaster_typeExtreme temperature	-0.072	2.274	-0.032	0.975
disaster_typeFlood	-0.593	0.373	-1.590	0.112
disaster_typeMass movement (wet)	-1.089	0.673	-1.619	0.106
disaster_typeStorm	0.000	0.282	0.001	0.999
disaster_typeVolcanic activity	0.107	1.311	0.081	0.935
disaster_typeWildfire	1.179	0.620	1.902	0.058
log(total_deaths)	0.161	0.065	2.460	0.014
log(no_injured)	0.422	0.058	7.306	0.000

2.2 Evaluating the log(no_affected) predictor -

After displaying the model coefficients and p-values, we realized that the coefficient for log(no_affected) was not statistically significant at all because of its p-value of 0.853, which is much greater than 0.1. We decided to explore one last model, a model with all the previous predictors as additive effects but without log(no_affected). We evaluated the performance of these models through another round of 10-fold cross-validation, comparing RMSE and r-squared to determine the better model.

- Model 1 (with log(no_affected)): **log(no_affected)** + log_duration_days + disaster_type + log(total_deaths) + log(no_injured)
- Model 3 (without log(no_affected)): log_duration_days + disaster_type + log(total_deaths) + log(no_injured)

```
# A tibble: 4 x 5
  model   .metric  mean     n std_err
  <chr>   <chr>   <dbl> <int>   <dbl>
1 Model 1 rmse    2.23     10  0.0287
2 Model 1 rsq     0.236    10  0.0310
3 Model 3 rmse    2.23     10  0.0289
4 Model 3 rsq     0.237    10  0.0311
```

After evaluating the performance of the two models through cross-validation, we discovered that the values for RMSE and r-squared were similar again for model 1 and model 3. Because model 3 is more parsimonious and performed slightly better than model 1 with a lower RMSE value of 2.236 compared to 2.240, we decided to choose the model without log(no_affected) as our final model.

3 Results

Final Model (Model 3):

term	estimate	std.error	statistic	p.value
(Intercept)	8.952	0.325	27.506	0.000
log_duration_days	0.287	0.106	2.718	0.007
disaster_typeExtreme temperature	-0.037	2.264	-0.016	0.987
disaster_typeFlood	-0.596	0.372	-1.601	0.110
disaster_typeMass movement (wet)	-1.103	0.668	-1.651	0.099
disaster_typeStorm	0.004	0.281	0.013	0.990
disaster_typeVolcanic activity	0.106	1.310	0.081	0.936
disaster_typeWildfire	1.161	0.612	1.897	0.058
log(total_deaths)	0.166	0.059	2.787	0.005
log(no_injured)	0.423	0.057	7.365	0.000

Performance of final model on test data set:

```
# A tibble: 1 x 3
  .metric .estimator .estimate
  <chr>   <chr>          <dbl>
1 rmse    standard        2.59
```

For the chosen model, the model diagnostics, including the linearity, equal variance, normality, independence, and multicollinearity (more detailed information/analysis found in the appendix under model diagnostics), were all satisfied and there were no influential points in the data set.

The final model has an RMSE of 2.592, meaning the predicted values of log total economic damage (in thousands of USD) deviate from the observed values by approximately 2.592, on average. **In other words, the predicted values for total economic damage deviate from the observed values by approximately $13,356**. This is only slightly higher than the average cross-validated RMSE from the training set (2.236), which is expected when evaluating on unseen data, suggesting the model generalizes well and is appropriate for this data.

The statistically significant predictors in the model provide meaningful insight into what drives economic damage from natural disasters:

1. no_injured:

   For each 1% increase in the number of people injured, the expected total economic damage increases by approximately 0.422%, holding all other variables constant, making it the strongest and most statistically significant predictor in the model.

2. no_deaths:

   For each 1% increase in total fatalities, the expected total economic damage increases by approximately 0.161%, holding all other variables constant, providing further evidence that human casualties are positively associated with economic losses.

3. duration_days:

   Disaster duration also plays a meaningful role: for each 1% increase in the duration of the disaster in days, the expected total economic damage increases by approximately 0.285%, holding all other variables constant, suggesting that longer-lasting disasters tend to cause progressively greater economic losses.

4. disaster_typeWildfire:

   Wildfires are expected to multiply total economic damage by a factor of 3.193 (e^1.161 = 3.193) compared to earthquakes (the baseline category), holding all other variables constant, though this result is only marginally significant (p = 0.058).

The coefficients from the model suggest that human impact measures, particularly the number of people injured and disaster duration, are the most important observable predictors of economic damage from natural disasters. The positive relationship between injuries and economic damage makes intuitive sense: a higher injury toll signals a more intense and widespread event, which simultaneously destroys physical infrastructure and strains public health systems — all of which translate into greater economic costs. The significance of disaster duration further supports this logic, as prolonged events allow for more cumulative destruction of assets and livelihoods.

4 Discussion

The most prominent finding from our project is that human impact measures (specifically the number of people injured and total fatalities) are the strongest predictors of economic damage. A 1% increase in injuries predicts approximately 0.422% higher economic losses, and a 1% increase in fatalities predicts approximately 0.161% higher losses, holding other variables constant. This aligns with our discussion in the introduction suggesting that casualties serve as a proxy for disaster intensity: more severe events damage infrastructure and harm public health systems, all of which result in greater economic losses. Disaster duration was also a meaningful predictor, with a 1% increase in duration predicting a 0.285% increase in expected damage, consistent with the intuition that prolonged events allow for greater cumulative destruction.

Notably, the number of people affected (no_affected) was not a useful predictor in our model ($p = 0.853$), which was somewhat unexpected given our initial hypothesis. We considered the possibility that its predictive value may also be largely captured by the injuries and fatalities variables already in the model, but this was disproven by a non-substantive VIF value ($x<10$).

Regarding disaster type, wildfires were the significant level as a predictor ($p = 0.058$), with expected damages roughly 3.25 times higher than earthquakes (the baseline), holding all else constant. While storms, floods, and earthquakes are the most frequent disaster types in our data, they are not meaningfully different predictors of economic damage from one another after controlling for human casualties and duration.

4.1 Limitations/Concerns

Some limitations should be noted. First, the EM-DAT database is subject to reporting biases: disasters in wealthier, more documented countries may have more complete damage estimates, while events in lower-income regions may be systematically under reported, reducing the reliability of our response variable. Also, the economic damage figures themselves are estimates and may reflect insurance payouts rather than true total losses, reflective measurement error. Finally, we removed a large number of 'N/A' data points; while we attempted to explore if the removed data points were systematically different in character than the rest of the data and didn't find anything, there could be hidden differences, causing the omission of the 'N/A data points' to render biases that we have not considered.

Furthermore, there are a considerable number of outliers in our data set with respect to total economic damage, duration in days, and many other variables. Although we applied the logarithmic function to those variables, these outliers should still be considered in the larger-picture analysis.

4.2 Practical Applications

The predictive findings from this analysis have meaningful implications for a lot of of public institutions. Government agencies and emergency management organizations could use an refined version of this model to generate rapid damage forecasts from early casualty estimates, informing how aid and recovery resources are prioritized in the immediate aftermath of a disaster. Infrastructure and city planning institutions could similarly use these findings to guide investments in disaster resilience since duration is a meaningful predictor of economic damage, which could yield great economic benefits.

Private institutions stand to benefit as well. Insurance companies and actuaries could account for injury counts, fatality projections, and disaster type into more data-driven risk pricing models, improving the accuracy of underwriting in regions where disasters are frequent. Hedge funds and banks could apply similar predictions to trading strategies involving catastrophe bonds, commodity markets, and real estate lending, where the ability to quickly estimate economic losses presents an edge in the market.

4.3 Building on Previous Work & Future Studies

Our findings build on previous work in this space. Noy (2009) showed that country-level factors like institutional quality and per capital income can significantly impact how severely a disaster disrupts a national economy. Our results are tangentially align with this: we find that human casualties and disaster duration are meaningful predictors of economic damage at the individual event level, suggesting that disaster intensity, not just country context, drives losses. However, where Noy operated at the macroeconomic level, our model works at the level of individual disaster events, which offers a more actionable predictive tool.

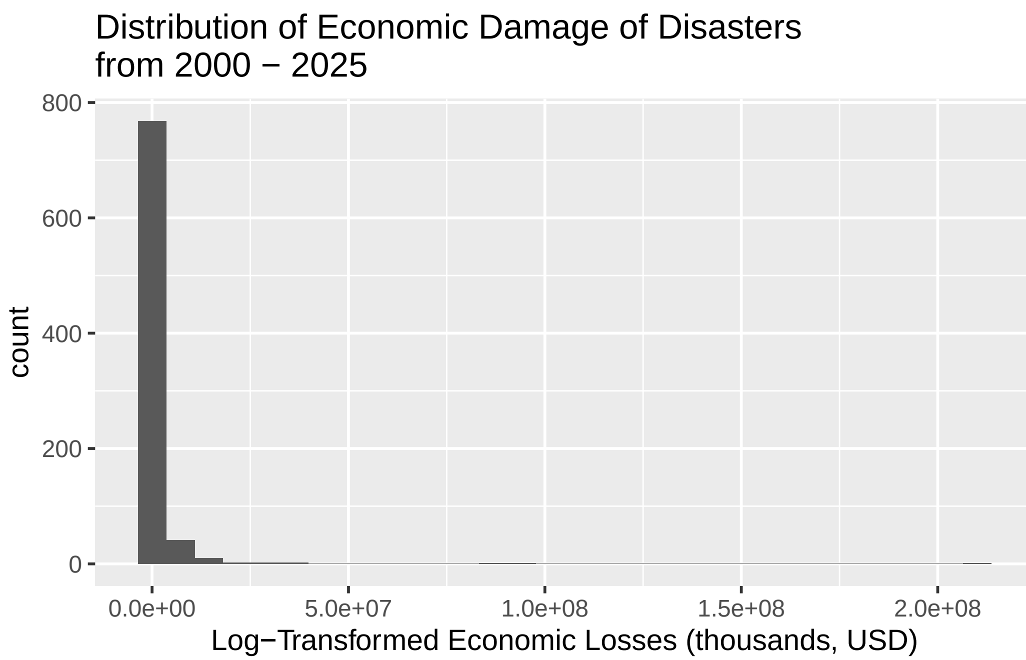
For future work, incorporating country-level (outside of the disasters themselves) predictors such as GDP per capita or an institutional quality index directly into our event-level model would be a natural extension that bridges both approaches, potentially capturing how the same disaster plays out very differently depending on where it strikes. Future analysis could also explore whether the predictive relationships identified here hold across different time periods given the documented increase in climate-related disaster frequency and severity. This may mean that older observations are less informative for predicting damages from more recent disasters.

Appendix

Exploratory Data Analysis

Response Variable: log of total economic damage in thousands of USD

This is the distribution of the response variable without the log transformation. The distribution is unimodal and extremely right-skewed. The center is best represented by the median, which is \$109,576,000, and the spread is best represented with the IQR, which is \$539,750,000 (559,750,000 - 20,000,000). There are 123 (828-705) outliers present in the upper end of our distribution.



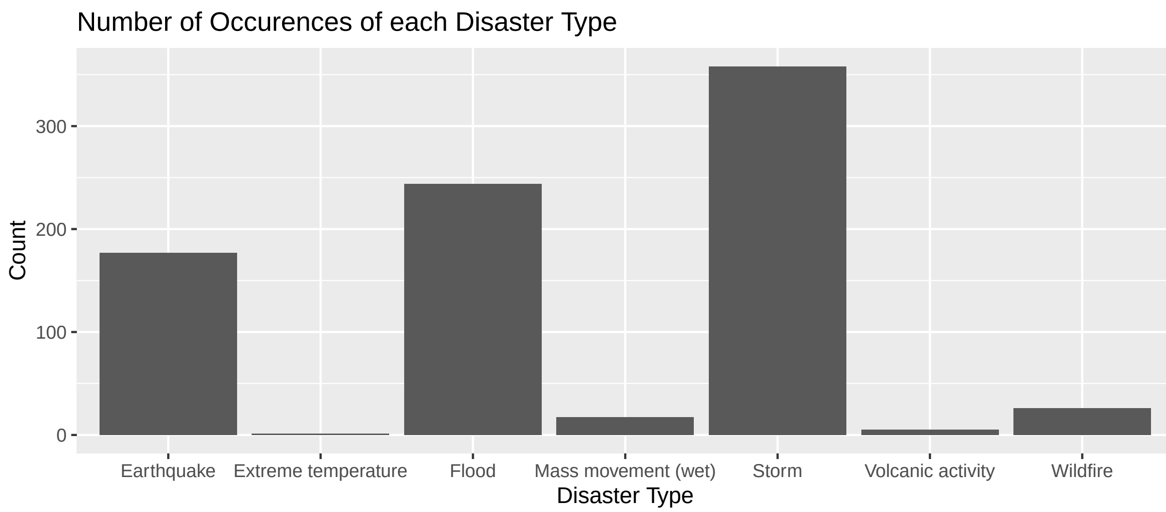
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12	20000	109576	1519107	559750	210000000

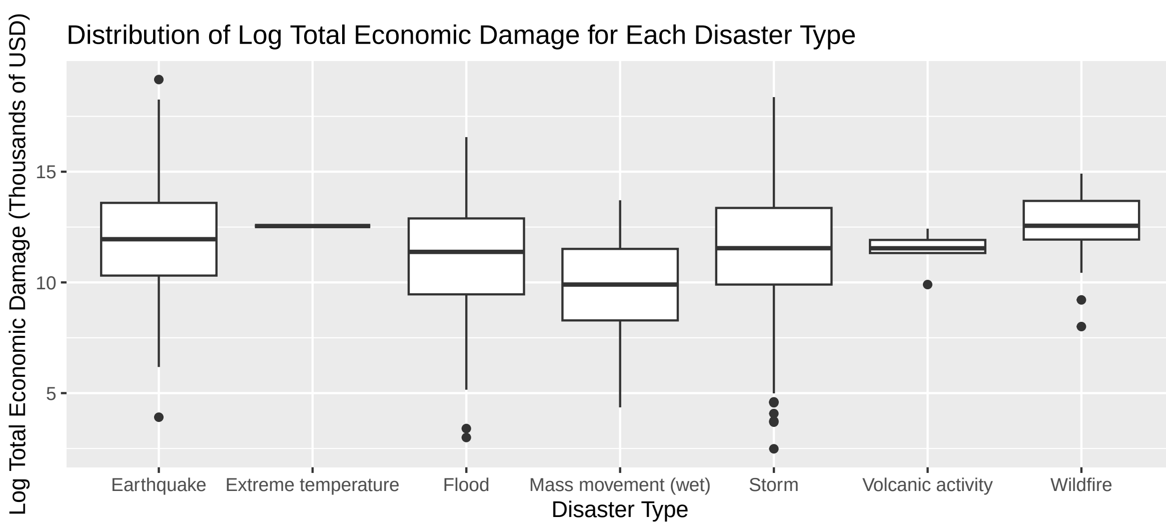
Predictor Variables:

1. disaster_type - the type of disaster (categorical)

Storms, floods, and earthquakes are the most common disaster types, composing most of the data. Storms are the most common. Extreme temperature, wet mass movements, volcanic activity, and wildfires are the least common disaster types, with extreme temperature being the least common.

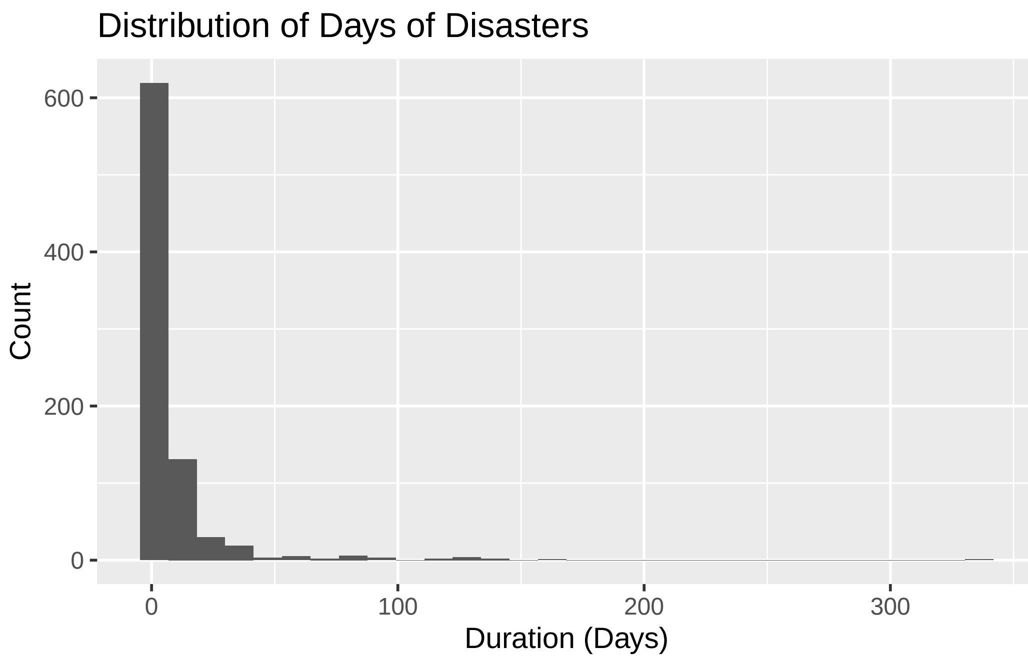
```
# A tibble: 7 x 2
  disaster_type            n
  <chr>                <int>
1 Earthquake             177
2 Extreme temperature      1
3 Flood                  244
4 Mass movement (wet)     17
5 Storm                  358
6 Volcanic activity        5
7 Wildfire                26
```

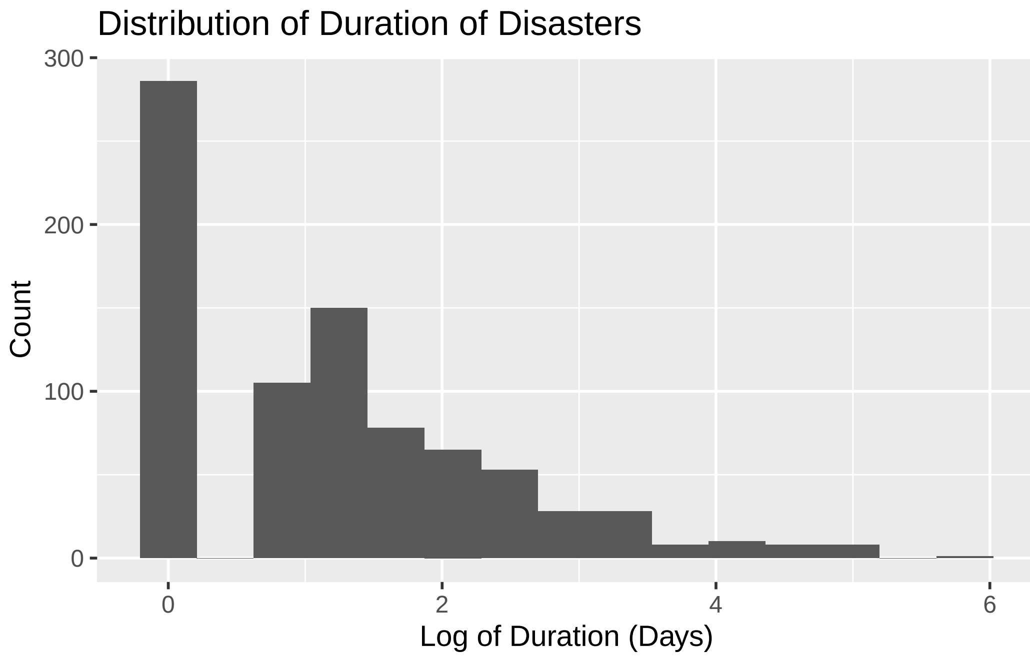




2. duration - duration in days of the disaster (quantitative)

Similar to the response variable, all of the quantitative predictor variables in our data set had extreme right skew because of the abundance of outliers. This motivated us to log-transform all of them. The histogram of the log of the duration has right skew and is bimodal. This makes sense because a lot of the disasters only last 1 day long due to their inherent nature, such as earthquakes. The median is 1.099 log duration in days, and the IQR is 1.946 log duration in days. There are outliers in the upper end of our distribution.



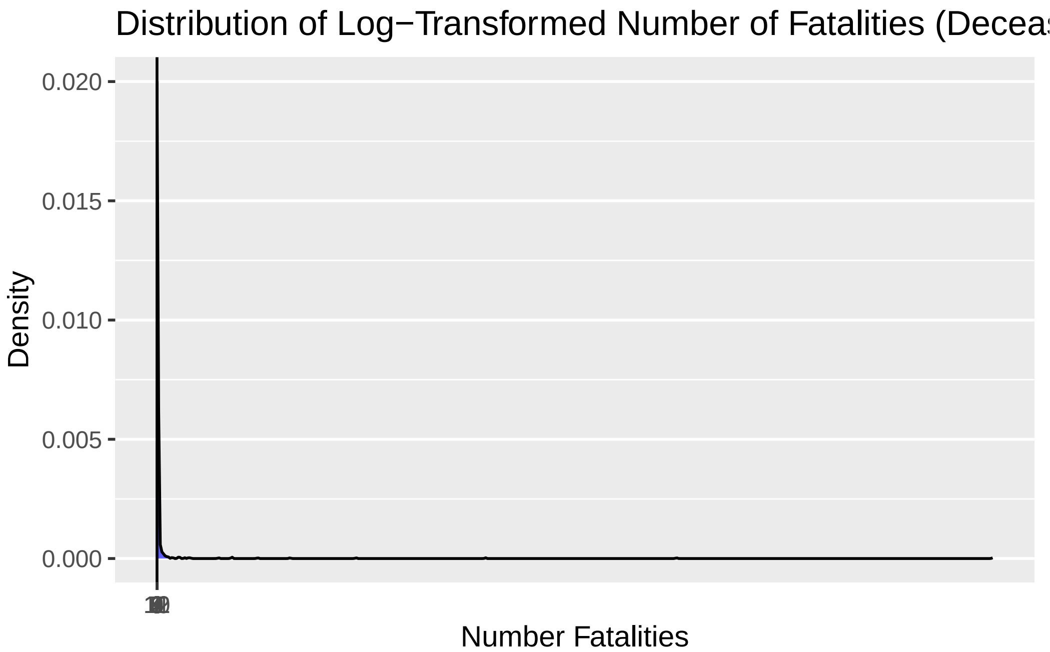


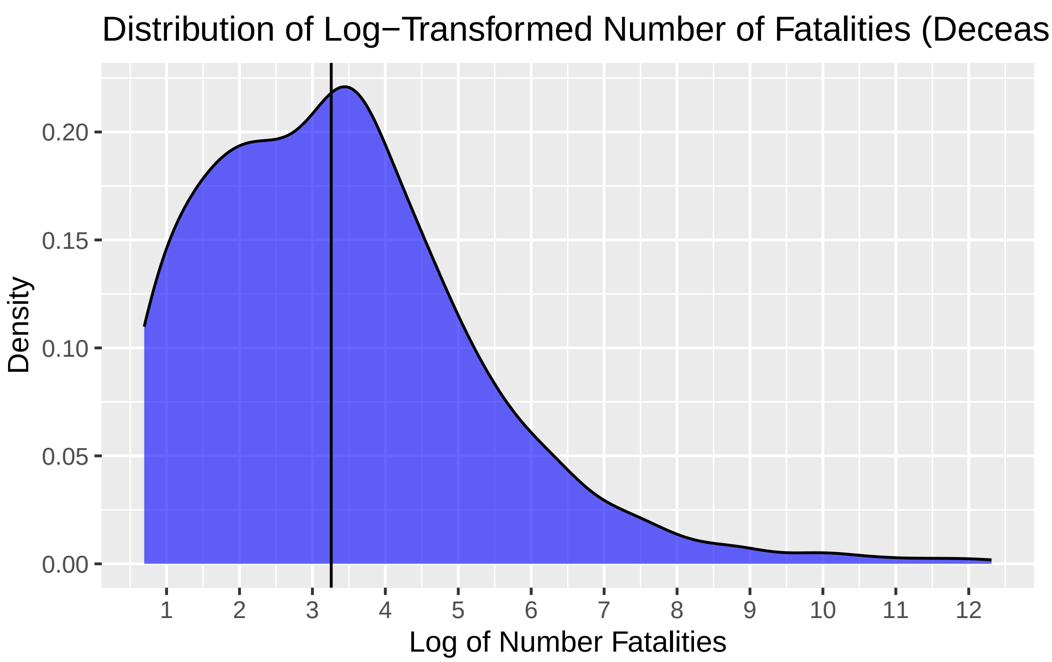

```
 Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.000   0.000   1.099   1.187   1.946   5.817
```

3. total_deaths - Total fatalities (deceased and missing combined) (quantitative)

   In the density plot above, we can see that the overall distribution of the log total fatalities is unimodal and has a distinct right skew. From the summary statistics, we can tell that while the median number of fatalities is 25 people, some grand disasters have a catastrophic number of deaths, with the highest number being 222570 people.

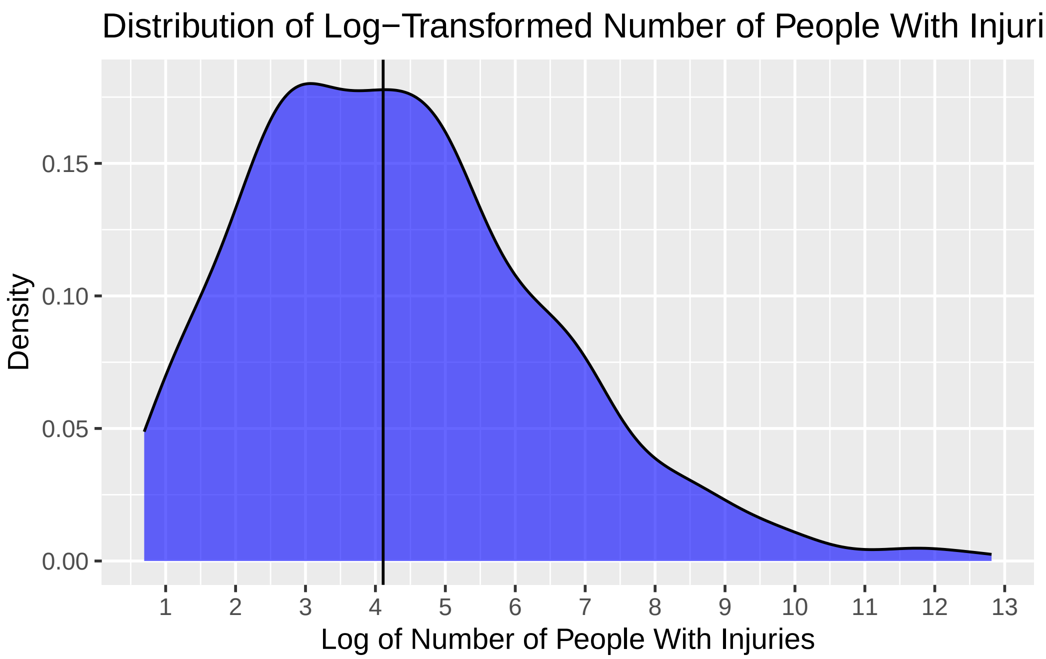
Statistic	Value
Min.	1.000
1st Qu.	6.000
Median	25.000
Mean	921.675
3rd Qu.	80.000
Max.	222570.000





4. no_injured - Number of people with physical injuries, trauma, or illness requiring immediate medical assistance due to the disaster (quantitative)

Statistic	Value
Min.	1.00
1st Qu.	14.00
Median	60.00
Mean	2338.36
3rd Qu.	266.25
Max.	366596.00

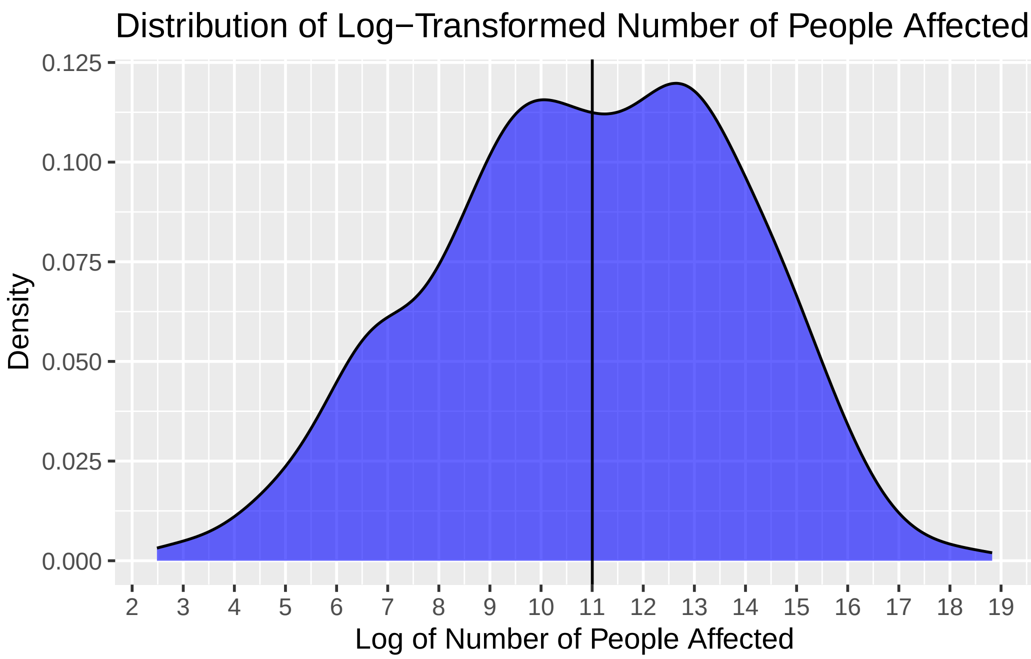


The story is similar to that of the previous predictor, log_total_deaths. There is a clear unimodal shape with a right skew, and we know that this skewness has been alleviated by the log transformation (so in reality, much larger). It makes sense that the median for log_no_injured is higher than the median for log_total_deaths at 60 people because logically there will always be more people injured than killed at any given natural disaster. This also applies to the maximum number being higher than that of total_deaths at 366,596 people.

5. no_affected - Number of people requiring immediate assistance due to the disaster (quantitative)

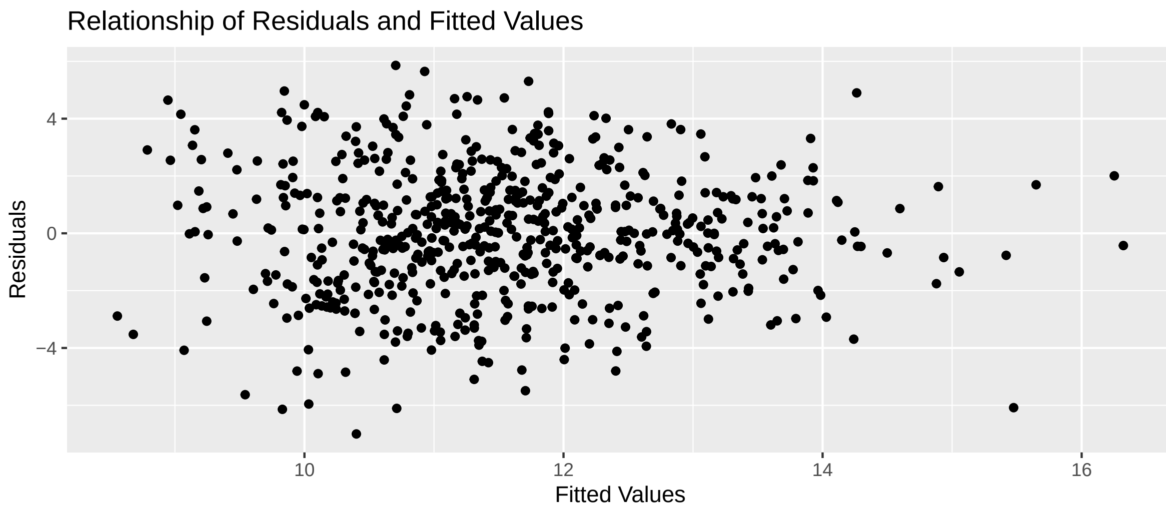
Statistic	Value
Min.	11
1st Qu.	7000

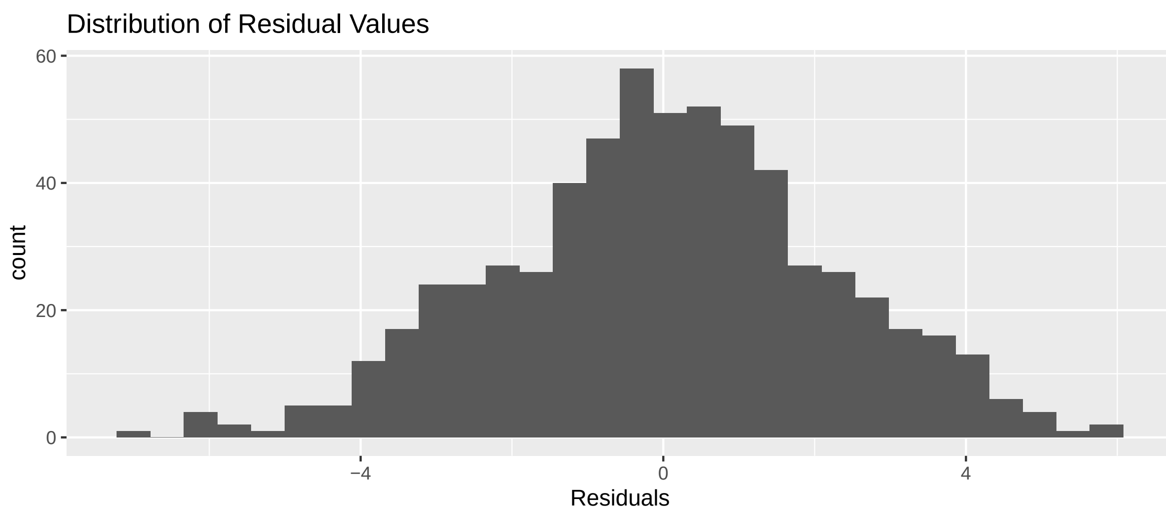
Statistic	Value
Median	60000
Mean	1325411
3rd Qu.	500250
Max.	150000000

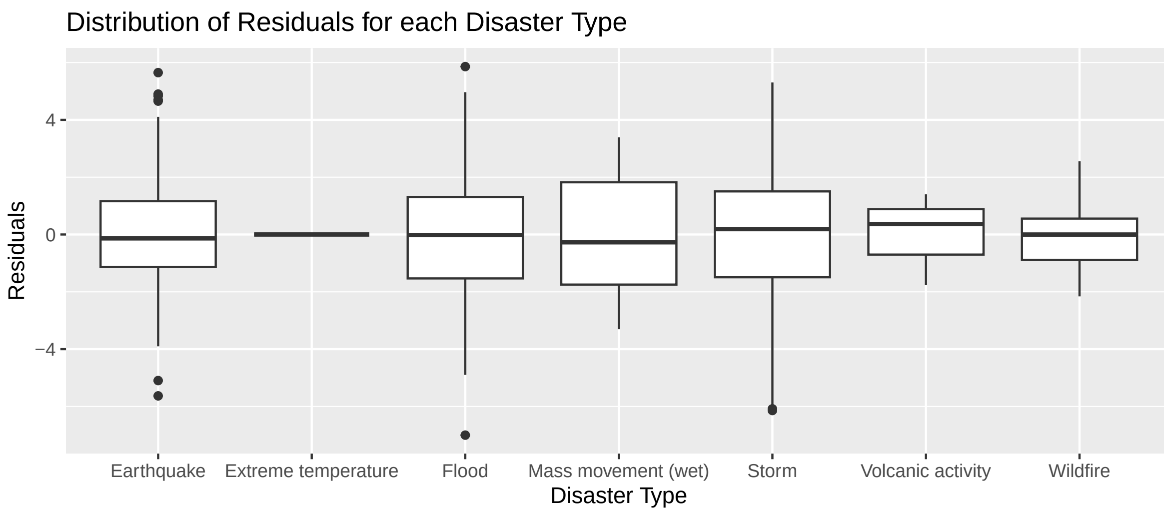


In the density plot above, the distribution of the log-transformed values appears roughly unimodal, though there is still a right skew. Compared to fatalities, the distribution of affected individuals is more spread out, reflecting the wide range of disaster scales – from local events to those impacting entire countries. From the summary statistics, there is a gap between the median (60,000 people) and the mean (over 1.3 million people), indicating a big right tail driven by a small number of extremely large disasters. The maximum value of 150,000,000 people affected highlights the scale at which certain disasters can disrupt entire populations. This makes sense considering the nature of natural disasters: while many events are relatively small, large-scale disasters can impact vast populations through displacement and infrastructure damage.

Model Diagnostics








```
# A tibble: 0 x 11
# i 11 variables: log_total_damage_000_us <dbl>, log_duration_days <dbl>,
#   disaster_type <chr>, log(total_deaths) <dbl>, log(no_injured) <dbl>,
#   .fitted <dbl>, .resid <dbl>, .hat <dbl>, .sigma <dbl>, .cooksd <dbl>,
#   .std.resid <dbl>
```

- The linearity condition is satisfied because the points are randomly scattered in the plot of fitted values vs. residual values. We can not guess whether the residual would be positive or negative for any predicted value.
- The constant variance condition is satisfied because the vertical spread is approximately equal for each predicted value. Although the cluster of points appear to form a fan shape, there are points outside of the fan shape pattern that allow this condition to be satisfied.
- The normality condition is satisfied because the distribution is unimodal and roughly symmetric and thus, similar to a normal distribution.
- The independence condition is satisfied because we are told that a disaster is included in the data set if at least one of the following criteria are met: 10+ people killed, 100+ people affected, declaration of a state of emergency, and solicitation of international assistance. Each disaster is an independent event as well. Furthermore, in the graph displaying the distribution of the residuals for each disaster type, we see that there is no correlation between the residuals and disaster type.

To determine if there is multicollinearity among the variables, we calculated the variance inflation factor (VIF) for each predictor variable.

	x
log_duration_days	1.959
disaster_typeExtreme temperature	1.037
disaster_typeFlood	3.555
disaster_typeMass movement (wet)	1.237
disaster_typeStorm	2.454
disaster_typeVolcanic activity	1.039
disaster_typeWildfire	1.398
log(total_deaths)	1.828
log(no_injured)	2.050

Because all of the VIF values are below 5, we concluded that multicollinearity is not prevalent among the variables.